

Combined exposure to polyvinyl chloride and polystyrene microplastics induces liver injury and perturbs gut microbial and serum metabolic homeostasis in mice

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ABSTRACT

A variety of microplastics (MPs) have become ubiquitous environmental pollutants, leading to inevitable human contact and health impacts. Most previous research has explored the toxic effects of a single type of MPs exposure. However, the effects of co-exposure to both common types of MPs, polyvinyl chloride (PVC) and polystyrene (PS) MPs on mammals have not been explored. Here, adult mice were exposed to PS-PVC (1.0 μ m PS and 2.0 μ m PVC both at the concentration of 0.5 mg/day) for 60 days. The results showed that PS-PVC co-exposure-induced hepatotoxicity was evidenced by liver histopathological changes, the release of inflammatory cytokines, and the activation of oxidative stress. Moreover, the intestinal mucosal barrier was damaged after PS-PVC treatment. The results of 16S rRNA gene sequencing reported there was a marked shift in the gut microbial structure accompanied by decreased relative abundances of probiotics, such as *Clostridium*, *Lachnospiraceae* UCG-006, *Desulfovibrio*, *Clostridiales* unclassified and *Ruminococcaceae* unclassified and increased the conditional pathogen abundances, such as *Erysipelatoclostridium*. Furthermore, the triglyceride (TG) and total cholesterol (TCH) expression levels in the serum and liver were increased after PS-PVC co-exposure. Serum metabolomics analysis showed that there were 717 differential expression metabolites found in the positive- and negative-ion modes, including 476 up-regulated and 241 down-regulated, mainly enriched in butyrate metabolism, thiamine metabolism, and phenylacetate metabolism. In addition, remarked changes in the gut microbiota and serum metabolic profiles were closely related to hepatic and intestinal injuries after PS-PVC co-exposure. These results have provided new insights into the toxic effects of PS and PVC MPs co-exposure through the gut-liver axis and the health risks of PS and PVC MPs should be paid more attention to humans.

1. Introduction

Plastic productions are indispensable in daily life due to their low cost, durability, and processability (Browne et al., 2011). The production and use of plastics have increased sharply. It is estimated that the global quantity of plastic products will reach 33 billion tons by 2050 (Phillips and Bonner, 2015; Prata et al., 2020). In recent years, the widespread existence of microplastics (MPs) has attracted more and more attention due to their potential toxicity to humans (MacLeod et al., 2021). Ingestion is considered to be the main source of human exposure to MPs (Xu et al., 2022). It is estimated that humans consume tens of thousands to millions of MPs every year, or a few milligrams per day

(Auguet et al., 2022). Many studies have detected MPs in food, air, drinking water, wastewater, and cosmetics (Auguet et al., 2022; Sangkham et al., 2022; Singh et al., 2022), revealing the possibility of MPs being ingested through the food chain (Oliveri Conti et al., 2020). According to previous reports, the content of MPs in drinks was 440 ± 275 particles per liter (Liu et al., 2021; Wright et al., 2020), while the content of MPs with small particle sizes is usually higher (Koelmans et al., 2019). A disposable paper cup with a plastic liner or plastic teabag has the ability to dispense billions of MPs and NPs into a cup of hot water (Ranjan et al., 2021). Previous research stated that the amount of MPs inhaled by humans from the air was approximately 3.5×10^5 – 2.2×10^7 particles, the number of MPs ingested from atmospheric deposition was

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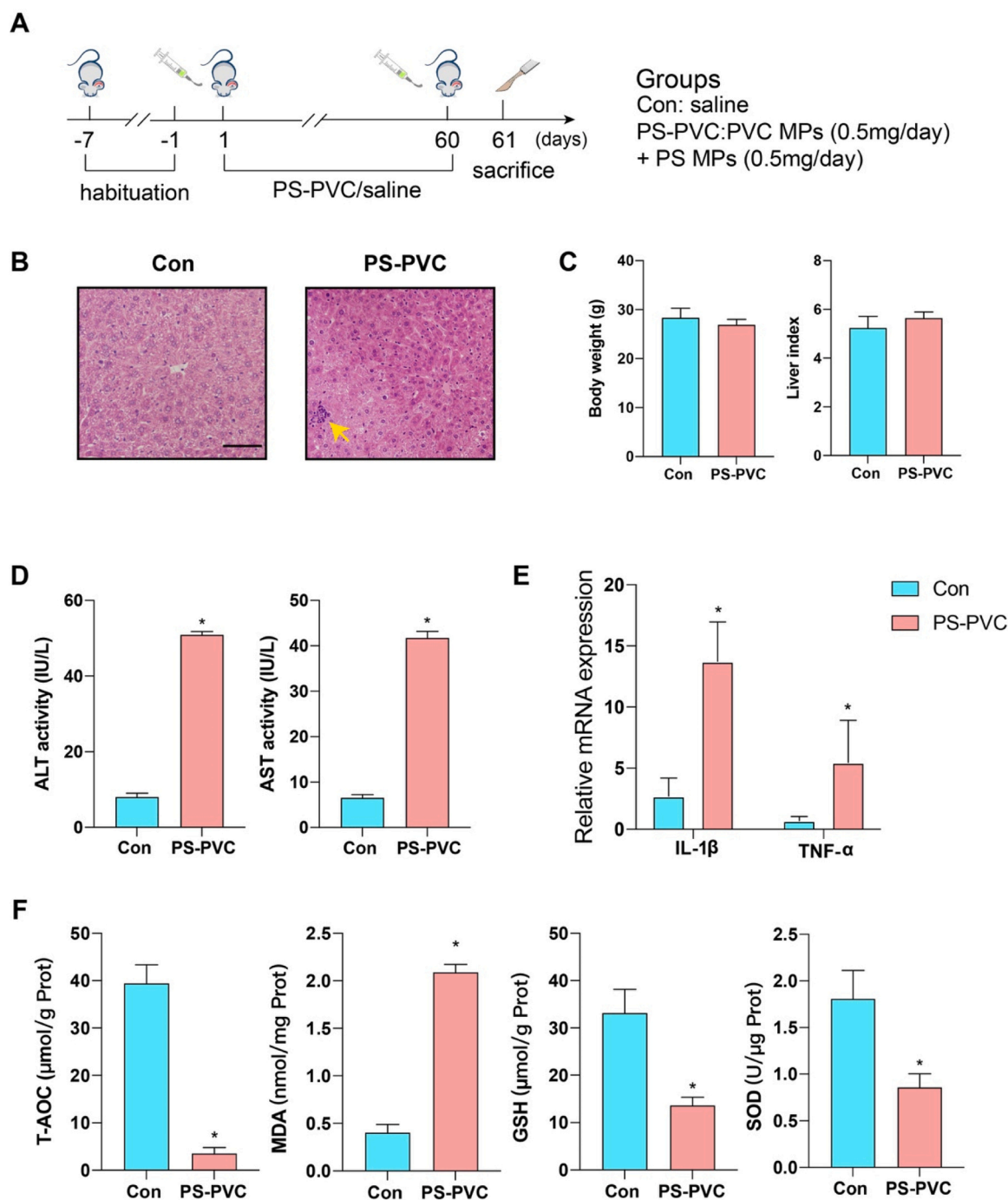


Fig. 1. PS and PVC co-exposure induced hepatotoxicity in adult mice. (A) Experimental schedule of PS-PVC exposure in adult mice. (B) Representative images of the liver by H&E staining (Scale bar: 200 μm). The yellow arrow represented inflammatory infiltration. (C) Body weight and liver coefficient. (D) ALT and AST expressions in serum. (E) The mRNA levels of TNF-α and IL-1β in the liver were measured by RT-qPCR. (F) The hepatic concentrations of T-AOC, SOD, GSH, and MDA (n = 4). Data represented as means ± SEM. * *p* < 0.05 vs. con group). Con, control group; PS-PVC, PS and PVC MPs co-exposure group.

approximately 1.9×10^5 – 1.3×10^6 particles, and the number of MPs consumed from food and beverages ranged from 0 to 1.337×10^4 (Fang et al., 2022). The first atmospheric measurement of MPs with large particle sizes, mainly composed of fibers, indicates that plastic particles are a related component of fine dust, such as the deposition rate of 575–1008 particles per square meter per day in central London (Wright et al., 2020). MPs are ubiquitous in the environment, and we need to be vigilant about the toxic effects of MPs on the human body. PVC and PS are the main types of MPs and more easily accessible to humans due to their wide applications, including toys, food and nonfood packaging,

shower curtains, medical tubings, numerous construction products, and electrical cables/wires (Bajt, 2021), and relatively low recovery, leading to underestimating their concentrations in sediment samples (Vermeiren et al., 2020).

So far, the accumulation of MPs and their potential toxicity to animals and humans have been occasionally reported. For example, MPs accumulation has been found in the human placenta (Ragusa et al., 2021), feces (Schwabl et al., 2019), blood (Leslie et al., 2022), and lung tissue (Jenner et al., 2022). In addition, exposed to larger polystyrene (PS, 40–100 μm) for 21 weeks, there was no obvious accumulation of PS

in the intestine or liver of mice, but the imbalance of gut microbiota, tissue inflammation, and disorder of plasma lipid metabolism could be observed (Deng et al., 2022). Acute exposure to polyethylene (PE) and polyester (PES) MPs (1 mg/L) could induce gastrointestinal and immune disorders in zebrafish (Medriano and Bae, 2022). Continuous exposure to PS MPs (18 and 180 µg/kg/day) for 90 days exacerbated cyclophosphamide-induced liver toxicity (Wen et al., 2022). The toxic impacts of MPs exposure were found to significantly increase with the co-presence of other pollutants (Rios-Fuster et al., 2023). The administration of MPs (5 µm, 0.1 mg/day) and Tributyltin chloride (100 ng/kg body weight/day) exposure alone and in combination could change the composition of gut microbiota and spectrum of bile acids, while the combined exposure to MPs and Tributyltin chloride weakened the toxic effects of MPs and Tributyltin chloride exposure alone (Jiang et al., 2021). Current research has focused on the hazards of a single type of MPs to the organism. However, in the ecosystem, animals and humans are frequently exposed to multiple types of MPs simultaneously. The impact of simultaneous exposure to two or more types of MPs on animals and humans remains unclear.

More and more evidence supports the argument that intestinal microbiome homeostasis is important in maintaining host health (Franzosa et al., 2015; Trevelline and Kohl, 2022). Intestinal epithelial differentiation, nutrient acquisition, and immune regulation depend on the homeostasis of intestinal microbiota (Kamada et al., 2013; Wan et al., 2021). However, intestinal homeostasis is closely related to intestinal microbiota, epithelial cells and intestinal immunity (Antonini et al., 2019; Martel et al., 2022). Lots of studies have shown that the imbalance of intestinal microflora impairs the integrity of the intestinal barrier and causes harmful substances or pathogens to cross the epithelial barrier, leading to intestinal dysfunction (Kudelka et al., 2020; Larabi et al., 2020). Emerging studies have tried to explain the toxicity mechanisms of MPs through intestinal microflora. Our previous study has found that polyvinyl chloride (PVC) MPs (2 µm, 0.5 mg/day) induced gut microbiota dysbiosis (Chen et al., 2022b). Two different sizes of PS MPs (0.5 and 50 µm) at the concentration of 1000 µg/L for 5 weeks could cause gut microbiome dysbiosis and hepatic lipid disorder in mice (Lu et al., 2018). However, recent research has primarily concentrated on a single type of MPs regarding the gut microbiota composition, with only a few studies investigating the functional alterations of intestinal microbiota following exposure to two or more types of MPs simultaneously (Xu et al., 2022; Zhang et al., 2021a).

The liver is one of the main target organs attacked by external stimuli, and its normal function maintains a close relationship with intestinal microflora (Tilg et al., 2022). The normal intestinal mucosal barrier limits the systemic transmission of microorganisms and toxins and allows nutrients to enter the circulation and reach the liver, but they can also produce factors leading to liver injury under pathological conditions (Duan et al., 2019). When the intestinal mucosal barrier is broken, bacterial products, especially lipopolysaccharide (LPS) from gram-negative bacteria, can cross epithelial cells into the portal vein and then be localized on the liver tissue (Wen et al., 2022). LPS can trigger the excretion of pro-inflammatory cytokines, thus initiating the pathogenesis of liver and natural colon tissue. Therefore, we speculate that the hepatotoxicity caused by exogenous substances may be amplified in individuals with intestinal microenvironment disorder. In the present study, it is assumed that two types of MPs (PS MPs and PVC MPs) co-exposure may affect intestinal microbiota, further affecting the gut-liver axis (the correlation between intestinal microbiota and liver metabolism), and leading to host hepatotoxicity.

2. Materials and methods

2.1. Animals treatment

Healthy male C57BL/6J mice (6–8 weeks old) were acquired from the Laboratory Animal Center of Southern Medical University (SMU,

Guangzhou, China). All animals received food and water ad libitum and were housed in three per cage on a 12 h-light/dark cycle with a temperature of 23 ± 2 °C. This study was carried out in accordance with the National Institutes of Health Guide for the Care and Use of Laboratory Animals of SMU. After one week of adaptation, twelve mice were randomly divided into two groups, which were named control and PS-PVC (PS and PVC MPs co-exposure) groups, respectively. The mice from the PS-PVC group received PVC (2 µm) and PS (1 µm) both at the concentration of 0.5 mg/day for 60 days with oral gavage once a day (Fig. 1A). The mice from the control group received equal volume saline. In the present study, the selection of exposure time and dose of PS-PVC is based on previous studies (Dessi et al., 2021; Zhang et al., 2021a). PS MPs used in this study were obtained from Zhongke Leiming Technology Co., LTD (Beijing, China) and PVC MPs from Huayu Technology Co., Ltd. (Guangzhou, China). The characteristics of PS MPs and PVC MPs are shown in Fig S1. Similar to our previous study (Chen et al., 2022b), the mice received FITC-dextran (500 mg/kg, oral gavage), and then blood samples of mice from the caudal vein were collected to measure intestinal permeability after PS-PVC exposure. Prior to slaughter, all mice were euthanized with pentobarbital anesthesia at a dosage of 50 mg/kg, and their blood, tissue, and cecal content were collected for sequent experiments.

2.2. Histopathology analysis

The gut and liver tissues were fixed in 4% paraformaldehyde at room temperature for 48 h. As previous research described (Chen et al., 2022a), sections (4–5 µm thickness) were stained with hematoxylin and eosin (H&E). As our previous study described (Chen et al., 2022b), intestine slices were then stained with alcian blue-periodic acid schiff (AB-PAS) solution (Solarbio, Beijing, China).

2.3. Biochemical parameter analyses

The expression levels of triglyceride (TG), total cholesterol (TCH), and total bile acid (TBA) in the liver and serum were measured by commercial kits following manufacturers' instructions. To explore PS-PVC-induced oxidative stress activation in mice, the total antioxidant capacity (T-AOC), malondialdehyde (MDA), glutathione peroxidase (GSH), and superoxide dismutase (SOD) expressions in the liver were determined using commercial kits, which were purchased from the Nanjing Jiancheng Institute of Biotechnology (Nanjing, China). The expression level of serum LPS in mice was determined by the corresponding ELISA kits and accompanying operating instructions (Cusabio, China).

2.4. Intestinal microbiota analysis

Similar to our previous study (Chen et al., 2022a), microbial DNA was isolated from the cecal contents by a commercial kit (D4015, Omega, Inc., USA). The 16S rRNA (V3-V4 variable regions) were amplified through PCR. Primers were 341 F: CCTACGGGNGGCWGCAG and 805 R: GACTACHVGGGTATCTAATCC, respectively. Sequencing libraries were performed by the NovaSeq PE250 platform (LC-Bio Technology Co., Ltd, China). Raw data were performed under specific filtering conditions to obtain high-quality clean tags according to the fqtrim (v0.94). The sequences were assigned to the same operational taxonomic units (OTUs) with a 97% similarity using USEARCH (Version 10.0). The resulting sequence data was analyzed using the OmicStudio tools at <https://www.omicstudio.cn>. Abundance of gut bacteria was calculated using a two-tailed Wilcoxon rank-sum test for two groups. The Venn diagram showed common or specific OTUs in each group. Principal component analysis (PCA) was used to reflect the distance between different groups of samples based on the Bray-Jaccard method at the OTU level. The linear discriminant analysis (LDA) effect size (LefSe) method was used to determine the communities or species

Table 1
Effects of PS and PVC MPs co-exposure on the lipid indexes in serum and liver.

Indexes	Con	PS-PVC
Serum TG (mmol/L)	0.216 ± 0.059	0.864 ± 0.051*
Serum TCH (mmol/L)	0.575 ± 0.010	0.835 ± 0.012*
Serum TBA (mmol/L)	4.651 ± 1.233	5.523 ± 1.446
Hepatic TG (mmol/g Prot)	0.013 ± 0.005	0.132 ± 0.037*
Hepatic TCH (mmol/g Prot)	0.031 ± 0.003	0.045 ± 0.001*
Hepatic TBA (umol/g Prot)	6.849 ± 0.250	11.768 ± 0.103*

The presented values are the means ± SEM (n = 4).

* $p < 0.05$ versus the control group.

contributing to sample partitioning. The gene functions of these sequences were analyzed by the Kyoto Encyclopedia of Genes and Genomes (KEGG).

2.5. RNA extraction and real-time quantitative PCR (RT-qPCR) analysis

The total RNA was isolated from the colon and liver tissues with TRIzol (Takara, Japan). Reverse transcription of total RNA was conducted using the HiScript II 1st Strand cDNA Synthesis Kit (Vazyme, China). The qRT-PCR was coupled with the SYBR Green system (Vazyme, China), and the gene expression was performed on a Light-Cycler® 96 System (Roche Life Science, Germany). The information of all primers are shown in Table S1.

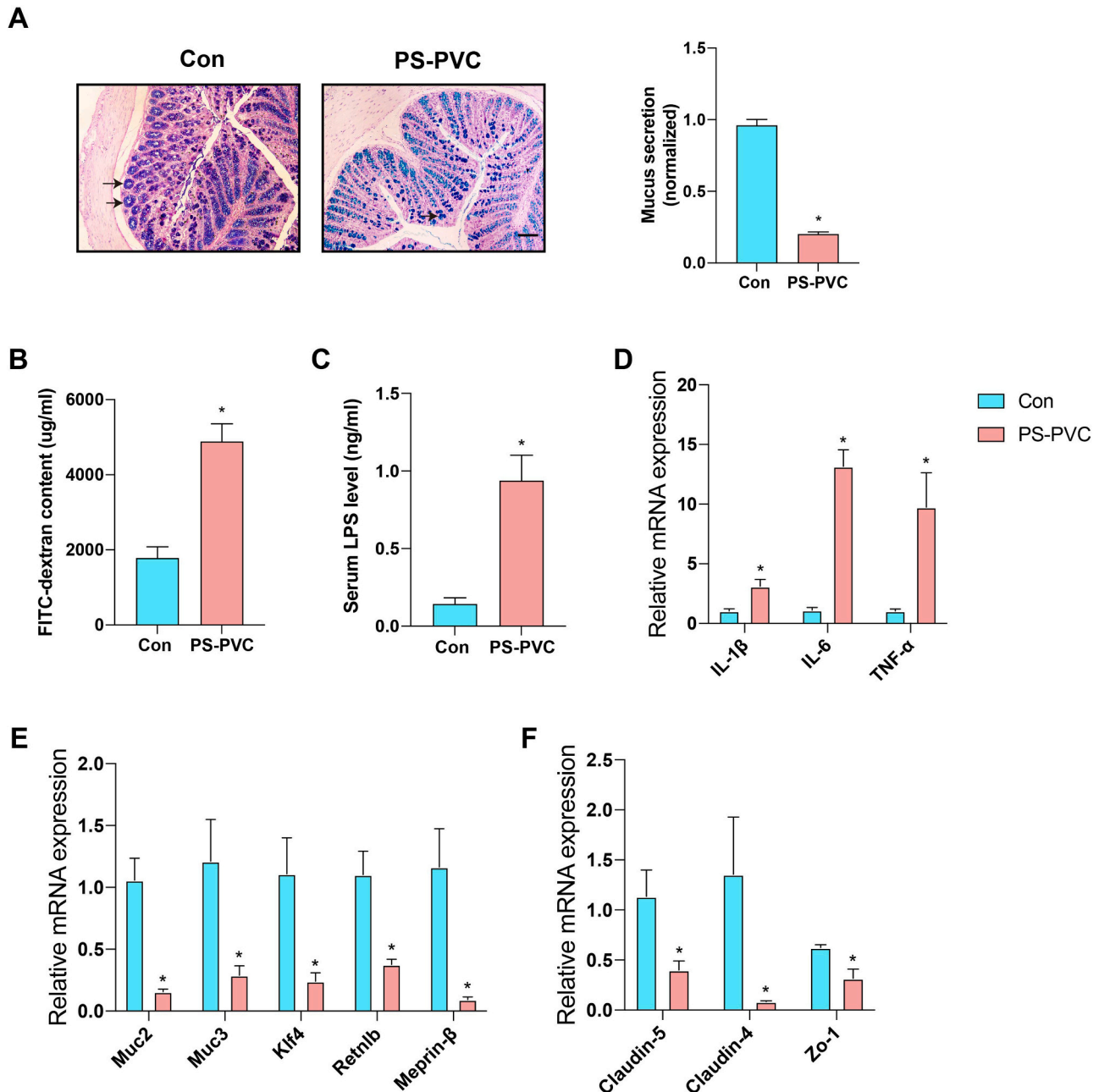


Fig. 2. PS and PVC co-exposure induced intestinal mucosal barrier disorder in adult mice. (A) Representative images of colon sections stained with AB-PAS (Scale bar: 200 μm). The arrow indicated the mucous compartment. (B) Colonic permeability analysis. (C) Serum LPS levels. (D) The mRNA expressions of IL-6, TNF-α, and IL-1β. The mRNA expressions of the mucus secretion-related genes (E), *Muc2*, *Muc3*, *Klf4*, *Retnib* and *Mep1b*, and tight junction-related genes (F) *Claudin-4*, *Claudin-5*, and *Zo-1* in colon tissue (n = 4). Data represented as means ± SEM. * $p < 0.05$ vs. control group). Con, control group; PS-PVC, PS and PVC MPs co-exposure group.

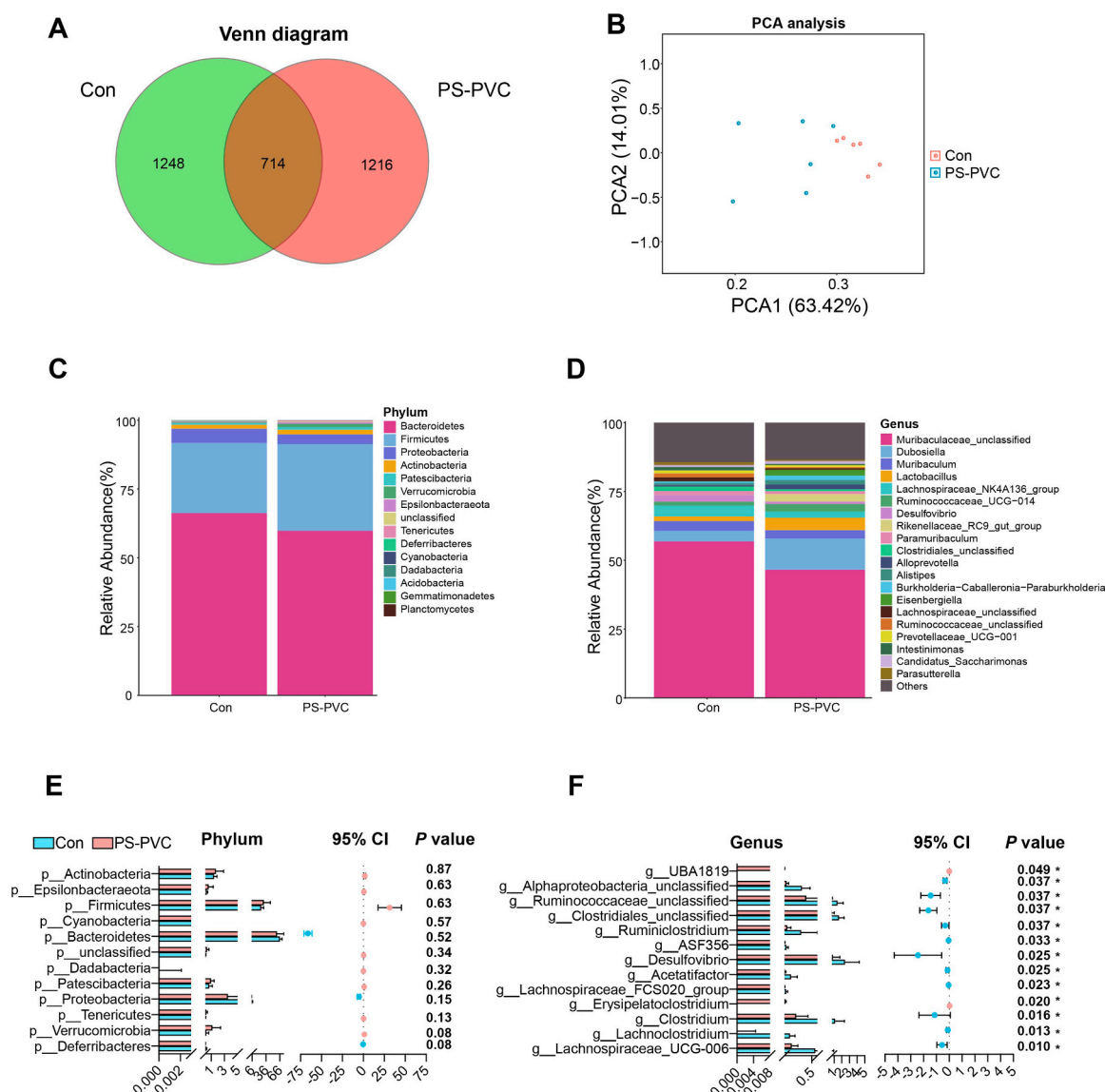


Fig. 3. PS and PVC co-exposure induced gut microbiota dysbiosis in adult mice. (A) A Venn diagram of two groups. (B) Principal component analysis (PCA) based on the Bray-Jaccard method at the OTU level. (C-F) Bacterial community compositions and the extended error bars at the phylum and genus levels between the two groups ($n = 6$). The data for each group are described as the means $\pm$ SEMs. Con, control group; PS-PVC, PS and PVC MPs co-exposure group.

2.6. Detection of untargeted metabolites in serum samples

As our previous study described (Chen et al., 2022b), a volume of 100 μ L of serum samples was performed with the liquid chromatography-mass spectrometry (LC-MS) analysis using a TripleTOF 5600 Plus high-resolution tandem mass spectrometer (SCIEX, Warrington, UK) with both positive and negative ion modes. The acquired LC-MS data pretreatment was performed using XCMS software and annotated with open-access databases, such as KEGG and HMDB.

2.7. Data analysis

All data are reported as the mean $\pm$ standard error of the mean (SEM) by the SPSS 20.0. Student's t -test was used to test the significance of differences between the two groups. A value of p less than 0.05 was thought statistical significance. Spearman correlation analysis was used to explore the relationship among the relative abundance of gut bacteria, biochemical indicators, and serum metabolites.

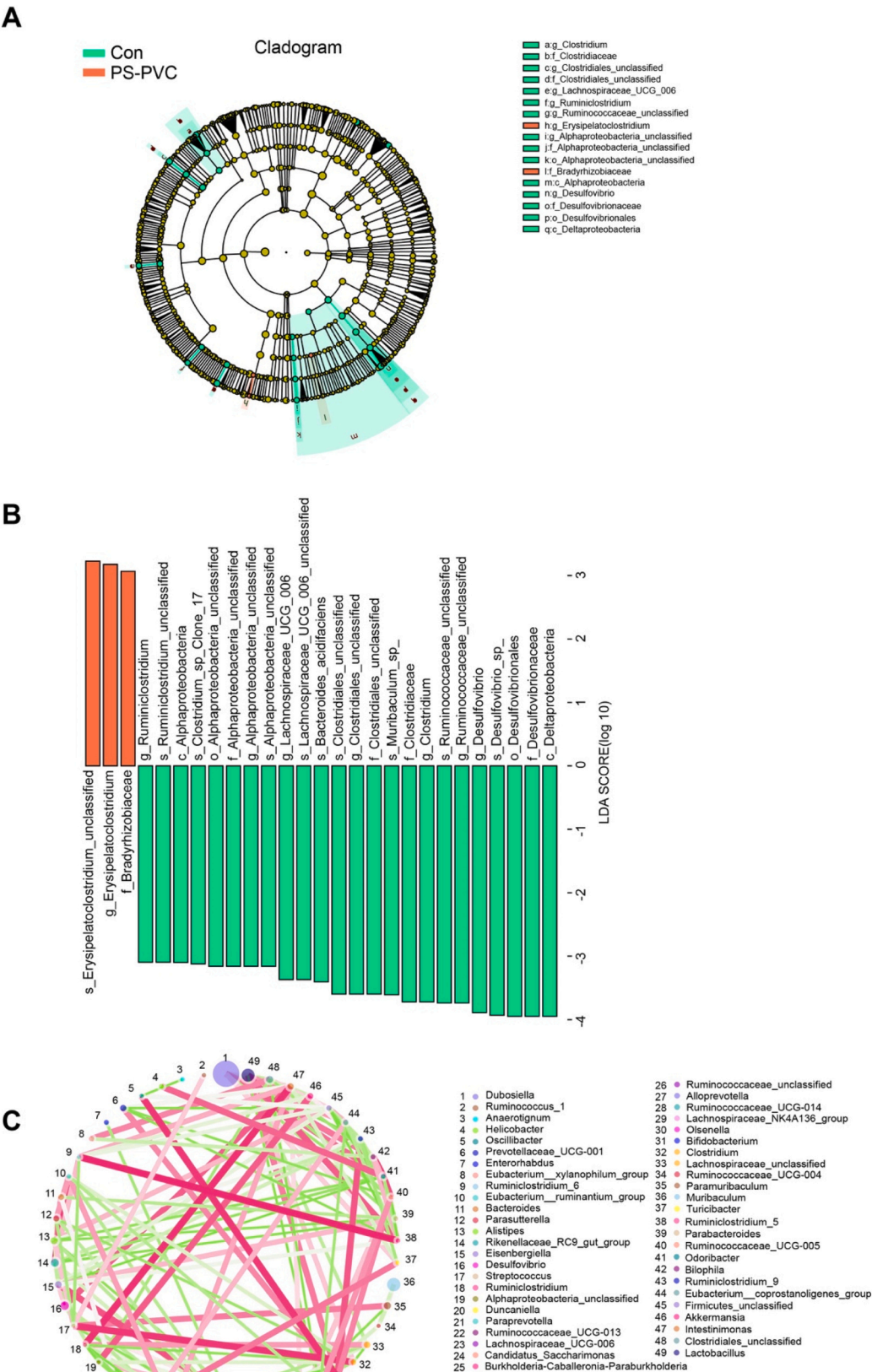
3. Results

3.1. Effects of PS-PVC co-exposure on the hepatotoxicity

Histological results showed that PS-PVC co-exposure induced liver inflammation, cell vacuolation, and karyopyknosis when compared to the control group (Fig. 1B). No statistical difference was observed in the body weight and liver index between the PS-PVC and control groups (Fig. 1C). Results of the biochemical analysis showed that PS-PVC co-exposure increased serum ALT and AST expression levels (Fig. 1D). The expression levels of TG, TCH and TBA were increased after PS-PVC co-exposure in serum and liver, except the serum TBA (Table 1). RT-qPCR results showed that TNF- α and IL-1 β were increased after PS-PVC co-exposure (Fig. 1E). In addition, PS-PVC co-exposure significantly decreased hepatic T-AOC, GSH, and SOD levels, and increased the hepatic MDA level (Fig. 1F).

3.2. Effects of PS-PVC on the gut injury

The AB-PAS staining result showed that the mucus secretion



(caption on next page)

Fig. 4. Differential biomarkers associated with PS and PVC co-exposure. (A) The phylogenetic distribution of differential taxa was visualized by cladogram. The circles from inside to outside in the cladogram indicate the phylum, class, order, family, and genus levels, respectively. The yellow circles represent the taxa with no significant differences. (B) LDA value distribution histogram reveals distinct taxa intestine microbiome composition in the PS-PVC and control groups ($n = 6$). The criterion of the differential taxa is LDA scores > 3 . (C) Correlation analysis of gut microbial community. Bacterial genera are represented by different colored circles and their abundances are positively correlated with circle diameter. The thickness of the line indicates the strength of the correlation. The red and green lines represent positive and negative correlations, respectively. Con, control group; PS-PVC, PS and PVC MPs co-exposure group.

remarkably decreased after PS-PVC co-exposure (Fig. 2A). The significant elevation of FITC-dextran concentration was observed in the blood after PS-PVC co-exposure (Fig. 2B). The serum LPS was increased after PS-PVC co-exposure (Fig. 2C). The result of RT-qPCR showed that the mRNA expressions of IL-6, TNF- α , and IL-1 β were increased after PS-PVC co-exposure (Fig. 2D). The transcription levels of colonic mucus secretion and tight junction (TJ) genes were measured and the results showed that PS-PVC decreased the mucus secretion-related genes, such as *Muc2*, *Muc3*, *Klf4*, *Retnlb* and *Mep1b*, and TJ related genes, such as *Claudin-4*, *Claudin-5* and *Zo-1*, expressions when compared to the control group (Fig. 2E-F). These results ultimately reflected gut barrier destruction after PS-PVC co-exposure.

3.3. Effects of PS-PVC on gut microbiota diversity and community

Through 16S rRNA sequencing analysis between the control group and PS-PVC groups, we explored the effect of PS-PVC on the composition of intestinal microflora. There were 1248 and 1216 unique OTUs and 714 common OTUs in both groups (Fig. 3A). As shown in Fig. S2, no statistical difference in alpha diversity was found in either group. The results of PCA indicated that at the OTU level, there was a difference between the two groups (Fig. 3B). As shown in Fig. 3C-D, the changes in the intestinal microbiota composition were found at the phylum and genus levels. In brief, *Bacteroidetes* and *Firmicutes* were the main phyla, followed by *Proteobacteria*, *Actinobacteria*, *Patescibacteria*, *Verrucomicrobia*, and *Epsilonbacteraeota*. *Dubosiella*, *Muribaculaceae_unclassified*, *Muribaculum*, *Lactobacillus*, and *Lachnospiraceae_NK4A136_group* were the main genera. PS-PVC co-exposure shifted the composition of intestinal microbiota. For example, when compared to the control group, lower abundances of *Bacteroidetes* and *Proteobacteria*, and higher abundances of *Firmicutes*, *Actinobacteria*, and *Patescibacteria* were found in the PS-PVC group, despite no statistical significance (Fig. 3E). At the genus level, PS-PVC co-exposure decreased *Lachnospiraceae_UCG-006*, *Clostridium*, *Desulfovibrio*, and *Ruminococcaceae_unclassified* but increased *Erysipelatoclostridium* abundances (Fig. 3F). The cladogram and linear discriminant analysis results indicated higher abundances of *Erysipelatoclostridium* and *Bradyrhizobiaceae* in the PS-PVC group when compared to the control group (Fig. 4A-B). Correlation network analysis (Fig. 4C) showed that *Prevotellaceae_UCG-001* was positively related to *Absiella* (0.6696), *Clostridiales_vadinBB60_group_unclassified* (0.7923), *Firmicutes_unclassified* (0.8671), *Ruminiclostridium_5* (0.7902), *Ruminococcaceae_UCG-005* (0.7773), but negatively related to *Dubosiella* (0.5385), *Facklamia* (0.6504), *Faecalibaculum* (0.6150), and *Oceanisphaera* (0.6792). *Dubosiella* was positively related to *Akkermansia* (0.7273), and *Lactobacillus* (0.6364), but negatively related to *Anaerotruncus* (0.8023), *Christensenellaceae_unclassified* (0.6969), *Clostridiales_unclassified* (0.7531), *Desulfovibrionaceae_unclassified* (0.7452), *Firmicutes_unclassified* (0.7762), *Mucispirillum* (0.7452). *Lactobacillus* was positively related to *Eubacterium_coprostanoligenes_group* (0.6035) and *Olsenella* (0.8169), but negatively related to *Anaerotruncus* (0.7880), *ASF356* (0.8228), *Papillibacter* (0.7631), and *Peptococcaceae_unclassified* (0.8658).

3.4. PS-PVC co-exposure disrupts serum metabolism

Serum samples from every mouse were collected and then measured the metabolites by LC-MS with positive-ion (Fig. 5) and negative-ion modes (Fig. S3). The results of PCA and orthogonal to partial least squares-discriminate analysis (OPLS-DA) plots ($R^2Y=0.9908$, Q^2

$=0.1269$) showed that there was a significant separation between two groups (Fig. 5A-B). There were 717 statistical differences ($VIP>1$, $p < 0.05$). Particularly, the volcano plot and heatmap showed the 476 up-regulated and 241 down-regulated metabolites in the positive-ion and negative-ion modes (Fig. 5C-D, Fig. S3C-D and Table S2), respectively. We also found that the following signal pathways were enriched in positive-ion mode, including thiamine metabolism, phenylacetate metabolism, phosphatidylethanolamine biosynthesis, alanine metabolism, and butyrate metabolism. While butyrate metabolism, thiamine metabolism, phenylacetate metabolism, D-arginine and D-ornithine metabolism, ketone body metabolism, and bile acid biosynthesis were enriched in the negative-ion mode.

3.5. Correlation analyses among gut microbiota, biochemical indicators, and serum metabolites

The analysis of Spearman's correlation was also constructed to explore relationships among the gut microbiota, differentially expressed metabolite profiles in serum, and biochemical indicators. As shown in Fig. 6, the result showed that there was a close correlation among gut microbiota, serum metabolites, and biochemical indicators. For instance, *Clostridiales_unclassified* displayed positive correlations with antioxidant indexes, like GSH (0.7381), SOD (0.7381), and T-AOC (0.7381), intestinal mucus genes, like *Klf4* (0.7381), *Mep1b* (0.7381), *Muc2* (0.7381), *Muc3* (0.7381), and *Retnlb* (0.7381), and TJ related genes, like *Claudin-4* (0.7381), *Claudin-5* (0.7381), and *Zo-1* (0.7381), and serum metabolites, like LysoPC20:5 (0.8095), but negative correlations with oxidant indexes, like MDA (0.7857), intestinal inflammatory cytokines, like colonic IL-6 (0.7857), colonic TNF- α (0.7857), and colonic IL-1 β (0.7545), hepatic indexes, like ALT (0.7857), AST (0.7857), hepatic TBA (0.7857), hepatic TCH (0.7857), and hepatic TG (0.7857), serum TG (0.7785), serum TCH (0.7857), and serum metabolites, like Phenacetic acid (0.7619) and 1H-Indole-3-propionic acid (0.7381).

4. Discussion

Recently research has shown that MPs can spread to humans through marine food, rice, and salt (Sangkhom et al., 2022; Zhao et al., 2016). Many studies have shown that there is a close relationship between intestinal homeostasis and the occurrence and progression of various diseases, and the liver is also considered to be directly suffering from intestinal discomfort (Febbraio and Karin, 2021). In order to assess the potential health risks of MPs, we try to simulate the actual intake of humans. In daily life, different kinds of MPs may be ingested at the same time. Compared with a single type of MPs ingestion from previous studies, the potentially harmful impact of multiple types of MPs ingestion on animals and humans is worth further exploration. The concentrations of PVC and PS MPs co-exposure are selected according to the actual environmental concentrations of MPs in rivers (Chen et al., 2022a; Eerkes-Medrano et al., 2015) and the level of human exposure through consumer goods (Dessi et al., 2021; Zhang et al., 2021a). In the present study, PS and PVC MPs co-exposure for a long time induced intestinal dysfunction and liver injury.

Many studies have shown that the toxicity of MPs is closely related to gut microbiota dysbiosis. For example, Angelo et al. reported that the size of 100 μm of PE and PES (both at the concentration of 1 mg/L) induced microbiota dysbiosis, metabolomic dysregulation, and oxidative stress in zebrafish by high throughput sequencing (Medrano and

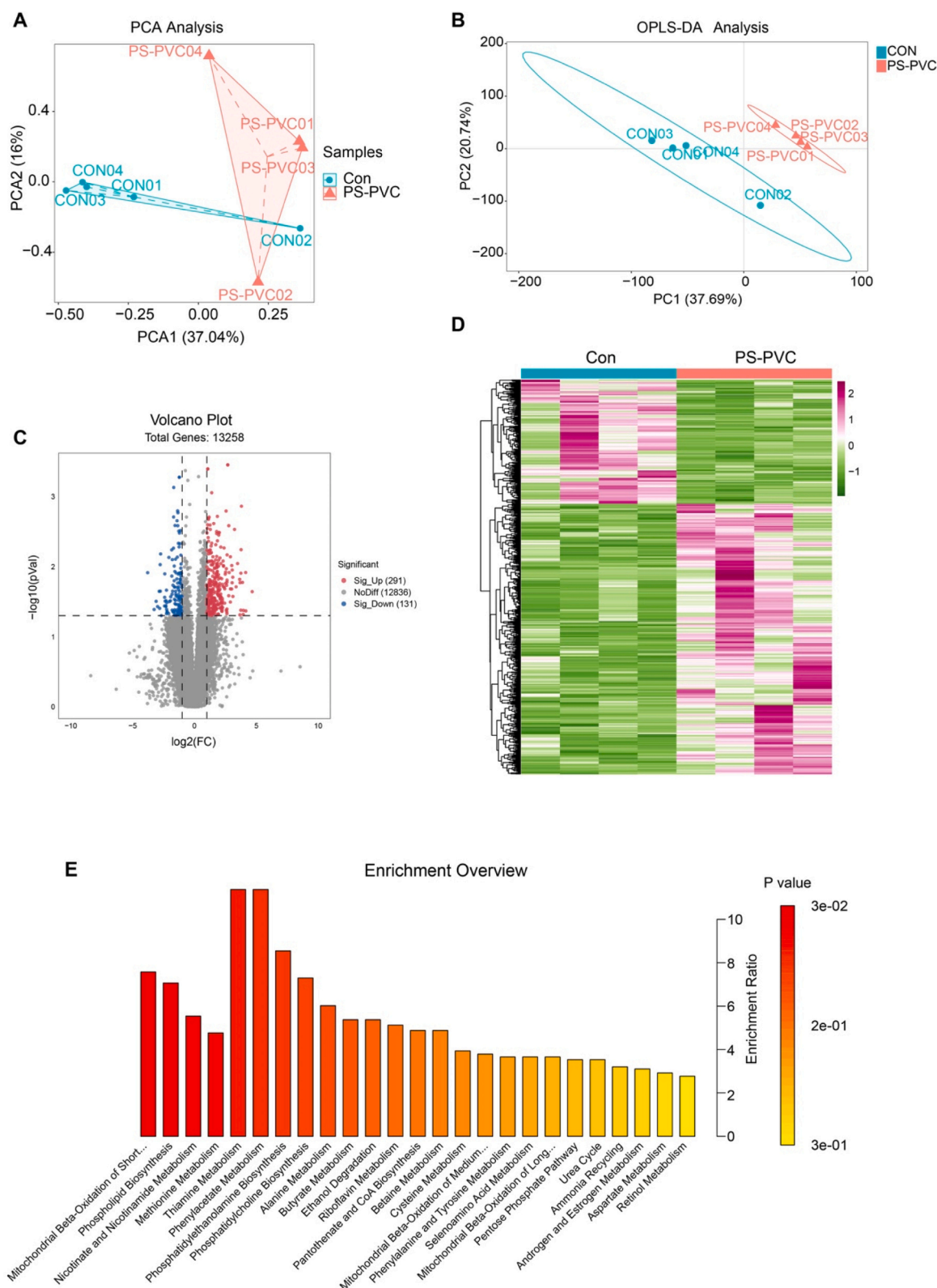


Fig. 5. PS and PVC co-exposure induced serum metabolomic changes in the positive-ion mode. (A) Principal component analysis (PCA) and (B) orthogonal projections to latent structures-discriminant analysis (OPLS-DA). (C) The volcano plot and (D) heatmap of differential metabolites. (E) Enrichment analysis of metabolic pathways in positive-ion mode ($n = 4$). Con, control group; PS-PVC, PS and PVC MPs co-exposure group.

Bae, 2022). Similarly, Lu et al. found that exposures to 1000 $\mu\text{g/L}$ of 0.5 and 50 μm PS modify the gut microbiota composition and induce hepatic lipid disorder in mice (Lu et al., 2018). Consistent with our previous study (Chen et al., 2022a), this work showed that PVC and PS

co-exposure for 60 days caused gut injury, manifested as intestinal permeability increase, intestinal mucus secretion decrease, and intestinal microbiota dysbiosis in adult mice. Previous studies have shown that PS MPs cause bacteria to invade epithelial cells by shifting the gut



Fig. 6. The heatmap of correlation analyses among the gut microbiota, biochemical indicators, and serum metabolites. The colors range from red (the positive correlation) to blue (the negative correlation).

microbiota composition, leading to gut mucosal barrier dysfunction (Jin et al., 2019). In the current study, we found that PS and PVC co-exposure decreased mucus secretion and transcription levels of several key genes related to mucus secretion genes (*Muc2*, *Muc3*, *Klf4*, *Retnib*, and *Mep1b*) and TJ-related genes (*Claudin-4*, *Claudin-5*, and *Zo-1*) expressions. The intestinal mucus secretion is the nutritional source of intestinal microbiota, and can also form an intestinal mucosal barrier to prevent the invasion of related pathogens (Desai et al., 2016; Paone and Cani, 2020). The imbalance of intestinal microbiota may damage the immune and barrier functions of the intestinal mucosa. In addition, some opportunistic pathogens may invade during the process of intestinal microbiota imbalance and pose a significant threat to host health (Wang et al., 2018). In the present work, we found gut microbiota community was changed after PS and PVC co-exposure. At the phylum level, PS and PVC co-exposure increased the *Firmicutes/Bacteroides* (F/B) ratio, which is suggestive of the metabolic disorder (Wen et al., 2022). In addition, PS and PVC co-exposure increased potential pro-inflammatory bacteria and conditional pathogens abundances, including *Rikenellaceae_RC9_gut_group*, *Alloprevotella*, and *Alistipes* (Chen et al., 2022b; Zhang et al., 2021b).

Meanwhile, PS and PVC co-exposure decreased several probiotics abundances, such as *Muribaculaceae_unclassified*, *Lachnospiraceae_NK4A136_group*, *Muribaculum*, *Prevotellaceae_UCG-001*, and *Intestinimonas*, related to short-chain fatty acid (SCFA) production and anti-inflammatory ability (Chen et al., 2022a; Chen et al., 2022b; Ma et al., 2020). We speculate that PS and PVC co-exposure may lead to intestinal microbial imbalance and dysfunction by changing the abundances of these bacteria.

More and more evidence shows that changes in intestinal microbiota are closely related to host inflammatory reactions, especially liver dysfunction. In our previous study (Chen et al., 2022a), PVC MPs (0.5 mg/day for 60 days) treatment induced corresponding mild liver damage in mice, which indicated that chronic intake of harmful substances might lead to the liver disease. Accordingly, in the present study, PS and PVC co-exposure (both at the concentration of 0.5 mg/day) in mice for 60 days led to more severe hepatotoxicity, which was manifested as histopathological damage, the release of inflammatory cytokines, and the activation of oxidative stress in mice's liver. The Spearman correlation analysis also found that the increased

hepatotoxicity was positively correlated with intestinal microbial dysbiosis caused by PS and PVC co-exposure. The destruction of the intestinal mucus barrier promotes the passage of bacterial products through intestinal epithelial cells into the portal vein, triggering the excretion of inflammatory cytokines, and leading to liver damage (Luissint et al., 2016; Wen et al., 2022). In this study, it was also observed that PS and PVC co-exposure increased serum and hepatic TG and TCH in mice, which was positively correlated with the abundance of *Erysipelatoclostridium* and negatively correlated with the abundance of *Clostridiales_unclassified*. The damage of BA homeostasis leads to liver cholestasis, which promotes liver fibrosis and inflammation (Baghdasaryan et al., 2016; Cai et al., 2022b). We found that co-exposure to PS and PVC increased the expression of TBA both in the liver and serum, although there was no statistical difference observed in the serum (Table 1). Intestinal BAs play an important role in enterohepatic circulation and are regulated by intestinal microflora (Cai et al., 2022a). Spearman correlation analysis indicated that *Erysipelatoclostridium* abundance was positively correlated with the BA expression. These findings suggest that the BA changes induced by PS and PVC co-exposure may closely be associated with gut microbiota dysbiosis. Herein, we also found the interaction between intestinal microbiota, metabolites, and hepatotoxicity, further expanding our understanding of PS and PVC co-exposure-induced hepatotoxicity and intestinal injury.

In addition, we try to explore the effects of PS and PVC exposure alone and in combination on animals based on our previous PVC MPs (Chen et al., 2022b), PS MPs exposures (Chen et al., 2023), and the present results. The results found that PS and PVC co-exposure seems to induce more severe hepatotoxicity and intestinal injury when compared to PVC exposure alone (Chen et al., 2022a) and PS exposure alone (Chen et al., 2023). There was a similar observation that TG, TCH, and TBA expressions were changed after PVC alone and PS-PVC co-exposure. While PS and PVC co-exposure induced more severe hepatotoxicity, displayed as the increased expressions of pro-inflammatory factors and oxidative damage, which was not found after PVC exposure alone. In addition, the permeability of the intestine was changed after PS exposure alone and PS and PVC MPs co-exposure. While we found that PS and PVC co-exposure decreased mucus secretion genes, including *Muc2*, *Muc3*, *Klf4*, *Retnlb*, and *Mep1b*, and TJ-related genes, including *Claudin-4*, *Claudin-5*, and *Zo-1* expressions and increased pro-inflammatory factors expressions, which were not found after PS exposure alone. Similarly, both PS and PVC treatment alone and in combination led to changes in the composition of gut microbiota.

In fact, there are still some shortcomings in the current research. There is evidence to suggest that MPs can adhere to other environmental pollutants, such as heavy metals, and endocrine disruptors (Xu et al., 2022). Therefore, the indirect toxic effects and potential mechanisms of MPs also need to be further elucidated. In addition to studying commercial MPs, more attention needs to be paid to the adverse effects of natural aging MPs with irregular shape, size, and composition on animals or humans. Currently, our previous studies were cited to compare the differences in the effects of PS and PVC treatment alone and in combination. In the future, further research is needed to explore the specific mechanisms that lead to this difference. Furthermore, the number of mice used in this study is limited, and more samples need to be added in the future to explore the toxic damage mechanisms of PS and PVC co-treatment.

5. Conclusion

In conclusion, this study shows that the combined exposure of PS and PVC MPs leads to liver dysfunction, lipid metabolism disorder, and activations of inflammatory reactions and oxidative stress in the liver. In addition, the combination exposure to these two kinds of MPs leads to the damage of the intestinal mucosal barrier, intestinal microbiota dysbiosis, and the change of the characteristics of serum metabolite, which provides new insights into the microbiota-gut-liver axis.

CRedit authorship contribution statement

Jingshen Zhuang: Methodology, Validation, Investigation, Founding acquisition, Writing – original draft, Writing – review & editing. **Qianling Chen:** Investigation, Validation, Writing – original draft. **Luyao Xu:** Investigation, Validation. **Xuebing Chen:** Conceptualization, Supervision, Writing – original draft and review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data needed to evaluate the conclusions in the paper are present in the paper and/or the [Supplementary Materials](#). Additional data are available from corresponding authors upon reasonable request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ecoenv.2023.115637](https://doi.org/10.1016/j.ecoenv.2023.115637).

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